

Unit 112 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 80.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 40 plots, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	214 sq ft/ac (SE 2.5 %)	Type	Sierran mixed conifer
Trees per acre	92	CWHR size / density	4 / D (cover 87 %)
QMD	20.6 in	SDI	279 (48 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	210 sq ft/ac 21.5 in	BA QMD	4 sq ft/ac 8.6 in
TPA	84	TPA	9
Volume	44.4 MBF/ac (8077 cu ft)	Volume	2.2 green tons/ac (72 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~156 sq ft/ac BA	Sampling error	5.1 % at 95 % (t = 2.02, n = 40)
Species (BA)	WF 66, PP 63, IC 36, DF 30, SP 19	Stratum standard	40 % - MEETS; 40 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	2	3.4	1.0	16	0.1
8-11.9	19	16.0	9.5	234	1.3
12-15.9	47	21.9	23.5	637	3.5
16-19.9	62	17.6	31.0	952	5.2
20-23.9	61	11.7	30.5	1068	5.9
24-27.9	69	9.5	34.5	1336	7.3
28-31.9	51	5.2	25.5	1066	5.9
32-35.9	42	3.4	21.0	926	5.1
36-39.9	28	1.8	14.0	680	3.7
40-43.9	12	0.6	6.0	296	1.6
44-47.9	12	0.5	6.0	307	1.7
48-51.9	9	0.3	4.5	247	1.4
52-55.9	9	0.3	4.5	248	1.4
56-59.9	1	0.0	0.5	26	0.1
60-63.9	4	0.1	2.0	111	0.6
All	428	92.5	214.0	8149	44.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	131	27.0	65.5	21.1	2522	13.9
PP	126	27.2	63.0	20.6	2383	13.1
IC	72	15.3	36.0	20.7	1415	7.8
DF	61	14.7	30.5	19.5	1130	6.2
SP	38	8.3	19.0	20.5	700	3.9
All	428	92.5	214.0	20.6	8149	44.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112: plots on the 300 ft grid (orange = QA flag)

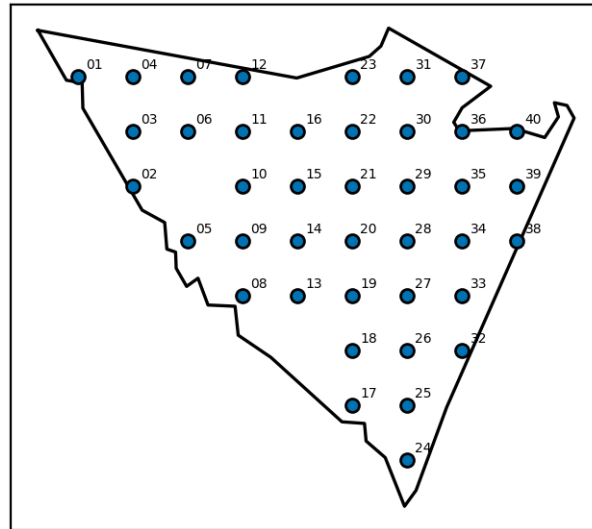


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
112-01	11	220	60	ok
112-02	11	220	24	ok
112-03	11	220	17	ok
112-04	12	240	30	ok
112-05	9	180	79	ok
112-06	12	240	147	ok
112-07	10	200	17	ok
112-08	5	100	112	ok
112-09	12	240	47	ok
112-10	11	220	43	ok
112-11	9	180	41	ok
112-12	11	220	129	ok
112-13	10	200	54	ok
112-14	13	260	21	ok
112-15	13	260	69	ok
112-16	13	260	72	ok
112-17	10	200	28	ok
112-18	12	240	105	ok
112-19	9	180	44	ok
112-20	11	220	98	ok
112-21	10	200	51	ok
112-22	13	260	33	ok
112-23	10	200	85	ok
112-24	13	260	96	ok
112-25	10	200	69	ok
112-26	12	240	61	ok
112-27	8	160	32	ok
112-28	10	200	44	ok
112-29	8	160	53	ok
112-30	11	220	26	ok
112-31	8	160	24	ok
112-32	12	240	77	ok
112-33	11	220	106	ok
112-34	11	220	68	ok
112-35	10	200	23	ok
112-36	13	260	186	ok
112-37	11	220	75	ok
112-38	9	180	72	ok
112-39	12	240	96	ok
112-40	11	220	114	ok